THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 4: **Algebraic Identities** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If $a - b = 5$ and $ab = 24$, then $a^2 + b^2 =$ 1

(a) 74 **(b)** 72 **(c)** 73 **(d)** 75

2. If $x + \frac{1}{x} = 4$, then $x^2 + \frac{1}{x^2} =$ 1

(a) 15 **(b)** 14 **(c)** 16 **(d)** 13

3. The value of 997^2 is: 1

(a) 994010 **(b)** 994009 **(c)** 994008 **(d)** 994011

4. The value of $93^2 - 57^2$ is: 1

(a) 5399 **(b)** 5401 **(c)** 5402 **(d)** 5400

5. Factorise $x^2 + x - 42$: 1

(a) $(x-6)(x-7)$ **(b)** $(x-6)(x+7)$ **(c)** $(x+6)(x-7)$ **(d)** $(x-5)(x+8)$

6. If $x + y = 14$ and $x - y = 6$, then $xy =$ 1

(a) 42 **(b)** 39 **(c)** 41 **(d)** 40

7. $(9x + 2y)^2 =$ 1

(a) $81x^2 + 36xy + 4y^2$ **(b)** $81x^2 + 18xy + 4y^2$ **(c)** $81x^2 - 36xy + 4y^2$ **(d)** $81x^2 + 4y^2$

8. $(2x - 3)(5x + 4) =$ 1

(a) $10x^2 - 6x - 12$ **(b)** $10x^2 + 23x - 12$ **(c)** $10x^2 - 7x - 12$ **(d)** $10x^2 - 7x + 12$

9. If $a - b = 2$ and $ab = 35$, then $a^2 + b^2 =$ 1

(a) 76 **(b)** 74 **(c)** 73 **(d)** 75

10. If $x + \frac{1}{x} = 5$, then $x^2 + \frac{1}{x^2} =$ 1

(a) 24 **(b)** 25 **(c)** 23 **(d)** 22

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. If $x + y = 13$ and $xy = 36$, find $x^2 + y^2$. 2

12. Find 998^2 using an identity. 2

13. Expand $(4x - 1)(2x + 5)$. 2

14. If $x + y = 15$ and $xy = 54$, find $x^2 + y^2$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. If $x + \frac{1}{x} = 9$, find $x^2 + \frac{1}{x^2}$ and $\left(x - \frac{1}{x}\right)^2$. 3

16. If $a - b = 3$ and $ab = 28$, find $a^2 + b^2$ and $(a + b)^2$. 3

17. Factorise $6x^2 + 7x + 2$. 3

OR

If $x + \frac{1}{x} = 10$, find $x^2 + \frac{1}{x^2}$ and $\left(x - \frac{1}{x}\right)^2$. 3

18. If $a - b = 8$ and $ab = 33$, find $a^2 + b^2$ and $(a + b)^2$. 3

OR

Factorise $4x^2 + 11x + 6$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Expand $(2x - 3y + 3z)^2$. 5

OR

If $a + b + c = 10$ and $ab + bc + ca = 31$, find $a^2 + b^2 + c^2$. 5

20. Expand $(2x - 3y + 4z)^2$. 5

OR

If $a + b + c = 15$ and $ab + bc + ca = 74$, find $a^2 + b^2 + c^2$. 5

END OF QUESTION PAPER — Total: 40 Marks

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